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IoT-based Cable Fault Detector with GSM and GPS Module using Arduino

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***Abstract*—The use of underground pipelines is increasing due to various advantages such as reduced pipeline losses, reduced maintenance costs, and increased resistance to harsh weather conditions. The measurement of distance displayed on the Liquid Crystal Display (LCD) varies whenever a system issue arises. There exists no subterranean cable which is greater than the previous approach because cables were meant to be positioned over the level of the head for a decade ago. Negative climate conditions, Storms, snow, torrential rains, and pollutants have no impact on under-ground lines, although it can be challenging to find a problem in an under-ground cable when it happens to a line. Underground cabling systems are more common in many urban areas. It is challenging to fix this specific cable even when the issue appears at that time because it is unclear where the cable failed because the problem is not clear. The work aims to solve this problem by proposing a suitable method for the digital age. This study has proposed a novel strategy based on Internet of Things (IOT) modules, Global System for Mobile Communications (GSM) and Global Positioning System (GPS). Arduino UNO, a module-based technique using the Google database is employed to identify faults. In this research work, the primary component is Arduino UNO. Through the router, this study has established a communication hotspot. Here, each MCU was connected to a transformer, and its status is checked by using the Google database. When compared to other procedures, the proposed method is more accurate and effective.**

Keywords—Cable fault detection, Global System for Mobile Communications (GSM), Global positioning system (GPS), Google database, Arduino Uno.

I. Introduction

Electricity has become a basic necessity in human lives. Most of our day-to-day appliances rely on electricity [1]. Since electricity has become so integrated into our way of life, it is crucial in every aspect of living. In the transmission and distribution of power, the transformer is a key piece of equipment. Underground cables are used in power networks to transfer electric current. The point, where then transmitted via above- and below-ground cables to the consumer ends [2]. Due to ageing and numerous flaws, underground cables must deal with a number of is-sues. Numerous research projects have been conducted to find solutions for these cable issues. Here, we offered a solution to these issues.

Numerous online and offline techniques are available to find faults and life in underground cables. The Murray loop approach: This method is frequently used to identify earth cable flaws. The Wheatstone bridge theory serves as the foundation for this test. By placing a Wheatstone bridge inside the earth cable, the fault spot can be located utilizing this examination. Varley loop technique Fourier transformation; Ohm's Law The problem must be reduced as quickly as possible to maintain the stability of the power network. This requires quickly functioning systems or procedures to immediately identify the issue so that supply distribution can be terminated in a short amount of duration. Because there are fewer approaches available, defect identification in underground cables is far more difficult than it is with overhead cables. The strategies we have studied in this research to reduce the numerous issues with underground cables are not particularly effective at identifying the issue [3].

Transferring electric energy from an outdated unit to consumers is the primary goal of electric transmission and distribution systems. Even though the under-ground connection framework offers superior dependability over the surface-to-surface links, discovering deficiencies is crucial for quality machines while in transit to fix them before they do further damage to the power device [4]. The process of finding problems has improved as a result of the constant need for support. With initiatives of sign preparation techniques and outcome-based procedures, developments of the shortcoming conclusion have been improved in recent years. The wavelet transformation has been shown to be useful for examining the transient sign produced by a high-caliber framework.

Therefore, this study has suggested a novel method to lessen the damage caused by faults in underground wires. And when compared to a variety of approaches, the accuracy of our suggested scheme is great. In this work, we've employed an IOT-based method with a Google database to expose faults with the aid of an Arduino WIFI module. It is entirely IOT-based. In comparison to other strategies, our suggested system has greater accuracy and efficiency.

II. Literature Review

A. Faults in Underground Cables

Conductor breakdown on one or more conductors results in these faults, often known as open circuits. These defects, which are frequently referred to as open circuits, are the result of conductor breakage on one or more conductors. The failure of cables and overhead lines at their joints, a circuit breaker failing in one or more phases, or a fuse or conductor melting in one or more phases are the most common causes of these issues [5]. The term "series fault" is also used to de-scribe open circuit problems.

B. *Fault for a short circuit*

When the insulation between the phase conductors, the earth, or both fail, a short circuit fault occurs. There are two types of faulty short circuits: Unsymmetrical fault and symmetrical fault.

C. *Fault Detection Methods*

• Online approach

In order to identify the failure spots, this approach uses and analyses the voltage and current samples. online techniques more uncommon than overhead lines for underground cable.

• Offline Approach:

This method entails using a particular tool to test the cable's field service. This offline strategy can be separated into two groups.
They are the Method of Tracer and Ternary method.

• Method of Tracer

By traversing the cable lines, the fault spot is located using this method. The audible signal indicates the fault. electromagnetic signal or signal. It is employed to locate faults very precisely.

• Ternary method

This technique allows you to find a cable issue without having to trace from either end. This method is used to pinpoint the overall location of the defect to speed up wire tracing.

This study proposes that, following the identification of a weakness inside the subterranean cable, Microcontroller remotely alerts the customer using IOT. The fault occurrence event up to a predetermined distance is routinely evaluated using the provided framework [6]. This methodology's fault finder measures the voltage and current of underground cable, and if there is a large discrepancy be-tween two issue identification, the locator will remotely alert the customer without approaching the problem. Today, a large majority of countries choose under-ground links for transmission over overhead links due to the numerous benefits that underground links have over overhead lines [7].

This activity aids in identifying the type of fault as well as its location. By delivering the information as a text message over GSM, we can determine the distance to the defect. Locating the line fault in the case of failure is the most challenging problem with underground wires. This activity aids in identifying the type of fault as well as its location. By using GSM, we may send the information as an SMS and get the fault distance. message format to the appropriate user through GSM. In this project, Ohm's law's basic principle is utilized. When a fault like a short-out happens, the voltage de-creases depending on how long the link problem has been there. An ADC is utilized to detect the voltage change and determine the defect distance. A link is made by connecting a number of resistors, and a power source located in the connection [8]. The defect distance is so discovered and displayed on the LCD. The issues with the underground cable network and their causes Because it is challenging to observe transmission, we must probe the ground and look at the complete network.

Because of this, a solution to this problem is required; as a result, this paper pro-poses a solution to this problem. In this approach, it is determined that the core idea of Ohm's law can be used to create a framework for identifying defect areas the length of the connection in constant temperature states, and the size of its cross-section is used to calculate the link's resistivity. The suggested framework identifies the precise location of the subterranean line breakdown [9]. The suggested system pinpoints the exact position of the issue. In response to the error is changed the new voltage applied to the ADC to deliver digital data for the programming device accurately. PIC IC also shows Isolation from Fault. The micro-controller displays the location, phase and time of the error on a 16x2 LCD. ESP8266 WIFI module is used in IOT to display data online. A web page containing information about the occurrence of the malfunction is generated and dis-played using HTML encoding [10].

Power wires used to supply energy are buried to prevent unwanted interference. It becomes quite difficult to pinpoint the precise position of the problems as a result. Numerous factors, including earthquakes, building, excavating, etc. can cause a fault to form. The line in question has a defect, but its location is unclear, making repair on that particular line difficult [11]. In this work, we show the operation of a hardware model that we developed ourselves to show how Ohm's law can be used to find defects in subterranean power lines. At regular intervals, it measures the voltage drop across the line. As soon as the issue appears, the volt-age drop across each interval alters in accordance with Ohm's law. An AT Mega 16 microcontroller has been used in the hardware model. The project employs a microcontroller kit to assess the exact position of a problem in subterranean. In metropolitan locations, the electrical cable flows underground rather than using overhead wires. Finding the precise location of an underground cable problem and repairing that particular line can be difficult. The fault is precisely located by the underground fault detecting system [12].

III. Components and Modules

A. *GSM Module*

Mobile modems are known as GSM or "Global System for Mobile Communications" (GSM). It is a form of mobile communication that is widely utilized around the globe. technology that is widely used throughout the world [13]. The GSM system was created using the Time Division Multiple Access (TDMA) method as a digital communication mechanism. Based on the implementation domain, every cell is different. On a GSM network, there are five main cell sizes: macro, micro, pico, and umbrella.

A GSM module can be used in a cable fault detector to assist in fixing cable issues live by providing real-time communication capabilities.

The GSM module can transmit data and information from the cable fault detector to a central monitoring station or a technician's device. This allows for remote monitoring of the cable's condition than any detected faults.

When the cable fault detector identifies a fault or issue in the cable network, it can send real-time data and alerts via the GSM module. This information can include fault location, fault type, and severity.

Many GSM modules include GPS functionality, which can help in pinpointing the exact location of the fault. This information is crucial for field technicians who need to reach the site quickly and accurately.

B. GPS Module

The GPS Module is used for navigating purpose. Only the module's position on the planet is confirmed, and. It generates data with its longitude and latitude and is one of a number of unaffiliated GPS devices that utilize the potent u-blox 6 positioning engine [14]. These versatile, inexpensive receivers have a compact enclosure and a variety of connecting options [9]. The tiny architecture, constrained power and memory options, cheap cost, and minimal space requirements of NEO-6 modules make them perfect for mobile gadgets that use batteries.

C. Relay

This tool includes a set of operational contact terminals and a set of enter terminals for one or more manipulate alerts [15]. Many contacts in various contact topologies, such as create contacts, break contacts, or combinations of these, may be present on the switch. Relays are utilized when more than one circuit has to be managed by a single signal or when a circuit needs to have its own, independent, low-power signal circuit in order to be refreshed. Electromagnets are typically used in relay designs. Other operational theories have been developed for solid-state relays, which control the connections with-out the need for moving parts thanks to semiconductor properties. Only one control power pulse is required by a latching relay to permanently operate the switch. A separate pair of control terminals or one with the opposite polarity must be pulsed in order to reset the switch because the same type of pulse has no impact and cannot be utilised again. Magnetic latching relays are helpful in circumstances when a power outage shouldn't influence the circuits.

D. LED

Light is produced when an electric current passes through a semiconductor light source known as (LED). Either a number of semi-conductors are used to create white light, or a layer of light-emitting phosphor is placed on the semiconductor device. Remote control circuits, which are present in many consumer electronics gadgets, use infrared LEDs. Early visible light emitting LEDs were ineffective and only produced red light. In the beginning, LEDs were often employed to replace tiny incandescent bulbs in seven-segment displays and as indicator lamps [11]. LEDs are used in these items as well as luminous wallpaper, auto headlights, advertisements, general illumination, traffic signals, horticulture traffic lights, and medical gadgets.

E. Embedded C Program

Liquid Crystal Display combines the solid and liquid phases of matter. LCDs use liquid crystals to create viewable images. When compared to Cathode Ray Tube (CRT) technology, displays using LCD technology can be significantly thinner. Polarized light is rotated by liquid crystals in an LCD display, which electronically turn on and off the pixels. Aside from being used in indoor and outdoor signs, LCDs are also used in LCD televisions, computer monitors, instrument panels, and cockpit displays. You may operate an LCD 16x2 in either 8-bit or 4-bit mode. Additionally, you can create your own characters. In addition to the 8 data lines.

F. Arduino Uno

The Arduino Uno is a microcontroller board that is open-source and designed by Arduino.cc relies on the ATmega328P MCU. It contains serial communication connectors, and some of them allow USB programming downloads (Universal Serial Bus).

G. Buzzer

Early devices used electromechanical technology that was identical to that of an electric bell, with the exception of the metal gong. In a manner similar to that, a relay may be programmed to stop the buzzing current that would otherwise activate it. These devices were typically mounted to a wall or ceiling to be used as a sounding board. The word "buzzer" was first used to describe the rasping sound that electromechanical buzzers produced.

Arduino is a widely adopted open-source programmable board, renowned for its user-friendliness and robust capabilities. It has found substantial popularity in both the hobbyist and professional domains. The core elements of Arduino include an Integrated Development Environment (IDE) for programming, with programs referred to as "sketches," and a microcontroller. Arduino boards are proficient at gathering various inputs, ranging from sensor data and button presses to online messages via platforms like Twitter, and translating them into corresponding outputs, such as motor activation, LED illumination, or online data publication.

H. ESP8266 Wi-Fi Module

ESP8266 is a tiny device makes it possible for microcontrollers to create simple TCP/IP connections with a Wi-Fi network links established using guidelines a la Hayes. However, originally, very little information about the chip and the orders it would receive was available in English. Because of the module's relatively low price and the possibility of potential mass production at incredibly low prices in addition to translating the Chinese documentation, many hackers were intrigued by the chip, software, and software on it.

I. Arduino Integrated Development Environment

IDE is used to create programs and upload them to Arduino boards. For developing, testing, and uploading code to an Arduino board, it offers a simple user inter-face. The Arduino IDE provides with a large selection of libraries and sample code to assist users in getting started with their projects. It supports a number of programming languages, including C and C++. The software also offers tools for debugging and testing code, as well as a simple interface for monitoring the code's output. Users may create and understand code more easily thanks to the Arduino IDE's code auto completion and syntax highlighting features. The widely used it is freely accessible to anyone who wants to use it for a range of projects and applications by professionals, students, and hobbyists.

IV. PROPOSED TECHNIQUE

The working block diagram and circuit of the proposed method is illustrated in Figure 1 and 2, respectively. The specific area of the deficiency will be identified in this essay. The study uses the conventional understanding of Ohm's law, in which low DC voltage is given feeder end through configuration where current changes depending on where cable inadequacy happens. According to the system's functioning, the current will fluctuate depending on how far the cable is from the location of the fault if there is a ground problem, or the three phases that precede it. when the voltage dips between the layout resistors change appropriately to generate The yield is then that is attached to the microcontroller, based on the issue conditions, after the microcontroller processes the advanced data.

Enhancing the effectiveness of the suggested approach can be achieved through the integration of cutting-edge technological components such as a microcontroller, sensors, and a display system. The sensor can pinpoint the precise location of cable faults, and this information can be instantly relayed to the user or worker in real-time, ensuring swift and immediate notification.

Optimal accuracy can be attained through regular sensor calibration. When high-quality and exceptionally dependable sensors are employed, they contribute to achieving precise results.

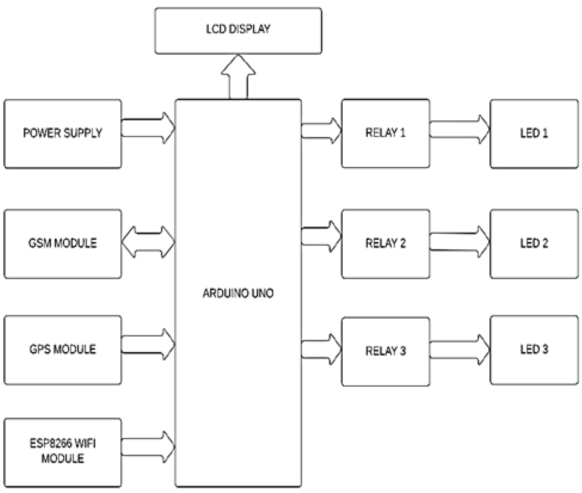

Fig 1. Block Diagram of the Proposed Design

Incorporating a GSM module into a cable fault detector serves the purpose of actively addressing cable issues by enabling instant communication capabilities. The GSM module is capable of relaying data and information from the cable fault detector to either a central monitoring station or a technician's device. This facilitates remote oversight of the cable's status and any identified faults. Upon detecting a fault or issue within the cable network, the cable fault detector can employ the GSM module to dispatch real-time data and alerts. These notifications encompass critical details such as fault location, fault type, and its degree of severity. Many GSM modules are equipped with GPS functionality, which plays a pivotal role in precisely determining the fault's exact location. This information proves indispensable for field technicians who require swift and accurate navigation to reach the site.

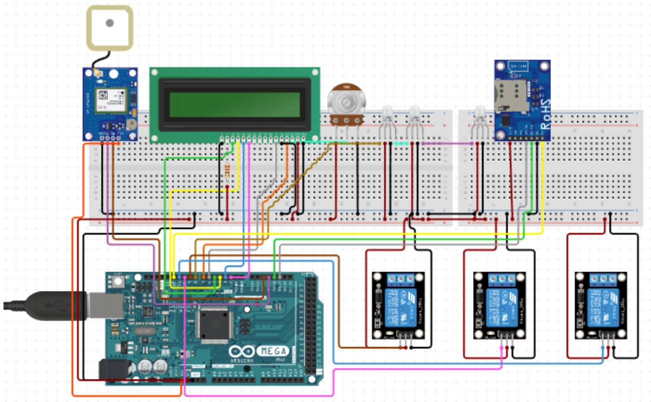

Fig 2. Circuit diagram of the proposed design

V. RESULTS

The selection of sensors for fault identification, along with additional sensors like the GPS module, GSM module, and Wi-Fi module, was made in accordance with specific criteria. As the project progresses, the critical factor demanding attention is ensuring the sensors' accuracy during their application.

With respect to the subterranean setting, damage caused by rats, etc., cables beneath the ground are vulnerable to a wide range of problems. Additionally, pinpointing the cause of a fault is challenging, and digging the entire line is re-quired to inspect it and correct any faults. Therefore, the proposed de-sign suggests a cable fault detection over IOT that determines the precise fault location over IOT and streamlines the maintenance process.

GPS-equipped cable fault detectors play a crucial role in monitoring and repairing large-scale cable networks. These devices utilize GPS technology to enhance the efficiency and accuracy of fault identification, location, and repair. In cases where an underground cable line experiences a fault, the process of repairing the cables can be notably challenging due to the absence of a reliable system for accurately pinpointing the fault's location. There is a clear need to develop a system capable of precisely identifying the location of faults that may occur in the cable network for all three phases - R, Y, and B - across various fault scenarios. According to Ohm's Law, it is evident that the cable's resistance is directly related to its length under consistent temperature and cross-sectional area conditions.

Eliminating the buzzing current necessitates identifying the fault location and then replacing the circuit or connection to establish a new pathway. The buzzing noise typically results from issues like loose wires, overloaded wiring, or improperly grounded wires.

In the process of cable fault location, the initial step involves testing the cable to detect any faults. Cable testing is typically the first phase of cable fault identification. During the testing, flash-overs are deliberately induced at vulnerable points within the cable, allowing for precise localization. Once the fault is identified, the location information is promptly transmitted to the relevant individuals' mobile devices through the integration of GSM and GPS modules for immediate response.

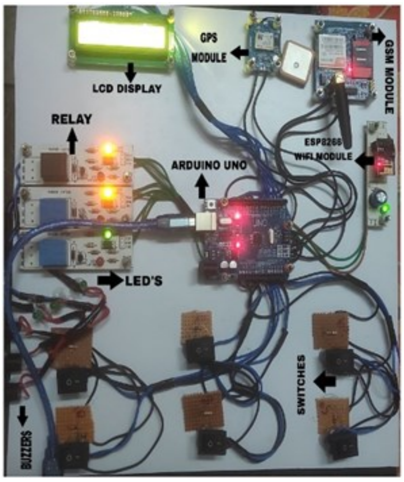

Fig 3. Implementation of the Proposed Design

The system relies on a potential divider network installed along the cable to identify faults. When a fault occurs, creating a short circuit between two lines, a distinct voltage is produced based on the configuration of the resistor network. This voltage is detected by the microcontroller and relayed to the user for real-time updates.

The advantages of precisely pinpointing faults through IoT include expeditious power system restoration, enhanced system performance, decreased operational costs, and reduced field fault location time.

The Arduino microcontroller works based on the output of the cable resistance. Relay helps to separate the faulty line from healthy line. The project detects only the location of the circuit fault in underground cable line, but it can also be extended to detect the location of an open circuit fault. The fault detected location is shared with the users with the help of GPS for immediate attention on fault by sending SMS via GSM module or by sending the information through Wi-Fi module.

IoT-based cable failure detectors can play a pivotal role in fortifying the resilience of cable networks and seamlessly integrating with pre-existing infrastructure, thereby curbing end-user downtime and service interruptions.

These detectors frequently leverage GPS and similar geolocation technologies to precisely identify the cable fault locations. This data is of immense value to maintenance teams, expediting fault resolution and, consequently, minimizing network downtime.

IoT-based cable failure detectors streamline operations, diminishing the need for continual manual monitoring and frequent on-site inspections. This leads to substantial cost reductions in network operations. By lessening the frequency and duration of service interruptions, cable networks gain heightened reliability, subsequently elevating customer satisfaction and trust in the service provider. IoT devices are designed with compatibility in mind, ensuring their ease of adoption and integration into the existing network management systems for a smooth transition.

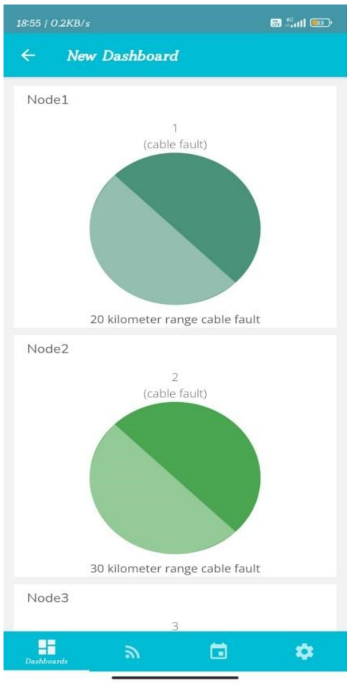

Fig 4. Fault Detection Displayed on UBIDOTS App

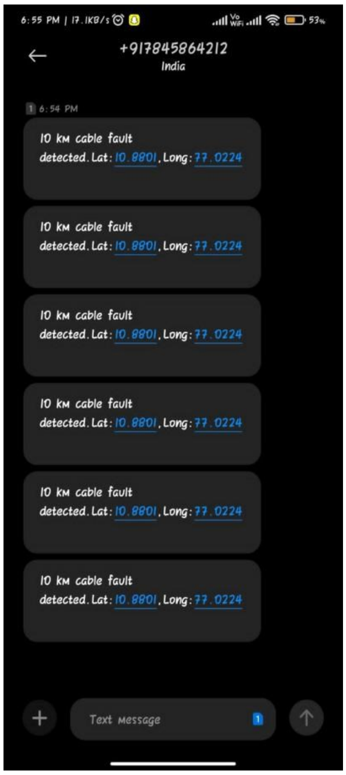

Fig 5. Fault Detection Displayed on Text message

Just that section needs to be excavated up to find the cause of the defect because the workers are aware of exactly which component is at fault. This enables more rapid service of underground cables while also saving a great deal of labour, cash, and hour. IOT technology is employed which enables the administration to track and ex-amine issues online. The potential divider network that has been built across the cable aids the system in fault detection.

A certain voltage is generated according to the resistor network configuration if a fault emerges at a spot where two lines are shorted altogether. The MCU detects this voltage and updates the user's screen. The user is informed of the distance that a certain voltage relates to. The MCU obtains the fault line information and exhibits it via an LCD. It also sends the information over the internet for online viewing. We developed an online platform that connects with the system to show cable issues online using IOT-Gecko. The implementation of the developed design is depicted in Figure 3. The fault detection displayed on UBIDOTS app is portrayed in Figure 4. Figure 5 illustrates the fault detection displayed as text message. Figure 6 shows fault detection displayed in LCD.

Fig 6. Fault Detection Displayed in LCD

VI. Conclusion

In the proposed model, the DC power supply, wires, controllers, and displays make up its four sections. Bridge-rectifiers convert AC signals to DC, rectifiers reduce 230V ac supplies, and regulators keep the voltage constant in the DC power supply portion. The cable portion can be distinguished using switches and a collection of resistors. The current-detecting section of the cable, represented a group of resistors and switches, is used as the originator to determine the defect at each point. Compo-nent determines by detecting the. The controlling component is then presented, ADC that transforms voltage input from the current detecting circuit into digital form. provides the signal to the microcontroller in digital form after signal con-version. The controlling unit's microcontroller, which is a part of it, performs the calculations required to figure out how far away the fault is, in order to guarantee a strong cable connection at every stage. The display component comprises of LCD screen coupled to a MCU that, in the event of a failure, displays the cable's state and distance at a given phase.

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